

Research article

A digital resource copyright protection scheme based on blockchain cross-chain technology

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ABSTRACT

With the expansion of the digital publishing industry, copyright infringement of digital resources has become increasingly rampant. Blockchain technology offers a promising solution for copyright protection of digital resources. However, existing copyright protection methods based on single-chain or consortium blockchains have disadvantages such as low system processing efficiency, poor scalability, and a lack of alignment with real-world business needs. To address the current challenges of blockchain technology in the field of copyright protection and establish a more effective solution for protecting digital resource copyrights, while also expanding the application of blockchain cross-chain technology and promoting its development, leveraging blockchain cross-chain theory, technology, and methods, a digital resource copyright protection scheme based on blockchain cross-chain technology was constructed. It outlines detailed processes for copyright registration, resource transactions, copyright modifications, and copyright enforcement. Furthermore, this copyright protection scheme was analyzed. The findings indicated that the blockchain cross-chain technology is suitable for digital resource copyright protection. The proposed copyright protection scheme can allow various stakeholders to better fulfill their roles, in addition to enhancing the overall maintainability and scalability of the blockchain system.

1. Introduction

According to the 52nd Statistical Report on Internet Development in China released by the China Internet Network Information Center, as of December 2023, with the Internet penetration rate of 77.5 %, the number of Internet users in China has reached 1 billion 92 million [1]. With the continuous increase in the number of Internet users, China's digital publishing industry has developed rapidly. The digital transformation of public libraries [2], characteristic resources of cultural centers, open classes, and novel media for cultural communication are becoming increasingly relevant. The digital networking and digital service capabilities of public libraries have improved considerably. According to the "Digital China Development Report (2020)," the number of registered users of the National Digital Library has exceeded 18.6 million. According to the Annual Report of China's Digital Publishing Industry, the revenue scale of the digital publishing industry in 2022 exceeded 1356 billion yuan, with a growth rate of 6.64 % compared with 2021 [3]. Overall, the digital publishing sector continues to grow rapidly, and its market size is expanding. However, numerous digital resource infringement incidents severely undermine the interests of digital resource copyright owners and render right protection difficult. For example,

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continuous piracy and infringement are highly prevalent in online literature. According to the 2021 China Online Literature Copyright Protection and Development Report, the scale of China's online literature piracy losses in 2021 was 6.2 billion yuan, indicating a year-on-year increase of 2.8 %. Literature piracy accounted for at least 17.3 % of the market share of the online literature industry, which severely damaged the original content ecology and affected the development of the online literature industry. To overcome the aforementioned problems, blockchain can be an effective tool. Blockchain technological convergence core technologies include consensus mechanism, distributed storage, cryptography, and smart contract, which exhibit decentralization, distribution, and tamper resistance [4,5]. By providing solutions for problems such as digital resource copyright protection and user rights incentive distribution, blockchain technology can provide the copyright governance of digital resources.

However, the current copyright protection methods based on blockchain technology are based on a single chain or consortium blockchains, which have the following shortcomings [6].

- (1) Low system processing efficiency: Copyright protection for all transactions is conducted on a single blockchain, leading to potential congestion and longer transaction confirmation times with an increase in the number of transactions. This adversely affects the system processing efficiency.
- (2) Misalignment with real-life transactions: Digital resource copyright protection involves multiple participating entities, with each node having to store data unrelated to its business. This makes it challenging to achieve a "division of labor" among participants, resulting in a situation that does not align with real-life transactions.
- (3) Poor scalability: New business nodes related to copyright protection can only be added to the original blockchain in a "sequential" manner, necessitating the re-backup of all data on the chain, which makes the expansion of the business difficult. Additionally, establishing effective cooperation between different business segments is challenging, which hinders the development of a robust ecosystem.

To address the aforementioned issues, this study explores a digital resource copyright protection scheme based on blockchain cross-chain technology. The present study has theoretical as well as practical significance, which can be explained as follows.

- (1) Theoretical significance: First, the study highlights the importance of cross-chain technology, providing theoretical support for interoperability between different blockchain networks. It also has theoretical implications for enabling digital resource exchange and collaboration between multiple blockchain networks. Second, the study introduces new ideas for digital resource copyright protection mechanisms, contributing to the innovation and evolution of theoretical frameworks for digital resource copyright protection.
- (2) Practical significance: First, the solution based on cross-chain technology expands the application scope of blockchain technology, promoting its development. Second, it offers practical guidance for digital resource copyright protection, enhancing the efficiency of implementing copyright protection in the digital resource domain.

2. Related research

2.1. Research on blockchain and copyright protection

Unlike conventional publication of paper-based works, digital resources exhibit simple mechanisms for information transmission and replication, and their role in the communication process can be changed, which complicates their copyright protection [7]. This is manifested in the lack of transparency in copyright transaction information, the difficulty of providing evidence in copyright infringement cases, challenges in rights verification, and limited profit sharing for creators. Blockchain technology, characterized by decentralization, distribution, and immutability, may be an effective solution for addressing these issues. Regarding the lack of transparency in copyright transaction information, a centralized digital resource trading platform does not provide ordinary users access to the details of the resource trading process, which results in severe information asymmetry among the trading platform, copyright owners, and ordinary users. Trading platforms have access to both digital resources and creator information as well as registered user information. Transactions in these platforms lack effective supervision paths and typically require final interpretation power over transaction data and processes, leading to low credibility of transaction information [8]. Second, difficulty in providing evidence for infringement and verifying rights protection. In the Internet environment, digital resource trading platforms cannot control the use of their resources on the Internet because of their centralized nature. Information on digital resources disseminates in a cheap and swift manner globally, as evidenced by the ubiquitous illegal sale and exploitation of others' works and information. However, for copyright owners, finding sufficient evidence for the infringement of their works is difficult [9], which is essential for the legal protection of interests. The process is cumbersome, complex, and highly expensive. However, defenders may not be able to obtain the desired compensation. Third, regarding low profit sharing among creators, the centralized nature of conventional digital resource trading platforms and the opacity of the trading process limit the reliability of transaction records. The profit-sharing model based on transaction records results in the loss of profits for digital resource creators [10]. Furthermore, most current Internet business models originate from the integration and processing of user data, and the same applies to digital resource trading platforms. A trading platform allows the collection of user information, behavior, transaction, and other data, which helps the platform gain considerable profits. However, in this process, a single user does not receive any profit sharing.

Chen [11] recognized the advantages of blockchain technology in areas such as peer-to-peer networks, consensus algorithms, and smart contracts. To enhance the efficiency of copyright management for ancient books in public libraries, the author constructed a

comprehensive framework for copyright protection in the digitization of ancient books in public libraries, leveraging the advantages of end-to-end management. Ji [12] attempted to address the prevalent issue of copyright infringement in the field of book copyrights by analyzing the role, channels, stages, and value flows of blockchain technology during the process of image rights confirmation and transactions. They proposed an extended value chain to assist platforms in enhancing the construction of a value system for protecting image copyrights through value co-creation. BELL [13] analyzed the drawbacks of blockchain in the copyright application process, highlighting the necessity for future legal regulations related to copyrights to address specific issues concerning data block storage, thus providing essential safeguards for the use of blockchain technology in digital copyrights. Li et al. [14] focused on achieving secure and efficient copyright authorization transactions to effectively protect original copyrighted works. Addressed issues in the consensus mechanism of blockchain technology, they proposed an improved DPOS consensus mechanism suitable for trustworthy copyright registration and authorization transaction scenarios. These studies acknowledge the value of blockchain technology in the field of copyright protection and provide guidance for copyright protection practices.

2.2. Research on blockchain cross-chain technology

In an endeavor to address the scalability issues caused by the "impossible triangle" theorem in blockchain systems [15] and against the backdrop of a growing number of blockchain projects, blockchain cross-chain technology has emerged and rapidly developed to facilitate interoperability among blockchain projects across different industries [16]. This enables the seamless flow of business and value between them. Currently, the development of cross-chain technology is in early stages, mainly including solutions such as the notary mechanism [17], sidechains/relays [18,19], hash locking [20], and mixing mechanism [21,22].

(1) Notary mechanism

The notary mechanism is a widely used and relatively easy-to-implement method. It introduces a trusted third party, known as a notary, as an intermediary between two blockchains that need to communicate across chains. The notary is responsible for collecting cross-chain data, confirming and verifying transactions, and ensuring the credibility and smooth execution of cross-chain transactions. Although the notary mechanism solves the problem of interoperability across chains, the reliability of the notary may not be ensured, posing risks.

(2) Side chains/relays

The side chain mechanism involves establishing a new chain, called the side chain, on top of the main chain, and transferring assets on the main chain to the side chain for trading and supervision. The results are then returned to the main chain. The side chain can have its own independent protocol and consensus mechanism, allowing efficient management of data without affecting the main chain's activities.

The relay mechanism is to build a relay network between different blockchain networks to process the transactions and data generated by them. The relay network can handle the interaction between different blockchains through smart contracts, and it can also have independent tokens and consensus mechanisms. Therefore, it can also be an independent blockchain, called the middle chain.

(3) Hash locking

The hash locking mode refers to a mechanism where users pay by guessing the original value of the hash value within a specified time period. In short, based on smart contracts, both parties first lock assets, and if both enter the correct original value of the hash value within a limited time, the transaction can be completed. The advantage of hash locking is that it has high security and flexibility, and it also supports cross-chain multiple transfers, making cross-chain operations more flexible.

(4) Mixing mechanism

The available mainstream cross-chain mechanisms have both advantages and disadvantages. For example, the notary mechanism that is easy to implement and widely used poses the risk of centralization; side chains/relays can expand the capacity of blockchains, but their transaction speed is slow; and hash locking is applicable to only asset exchange scenarios. To circumvent the shortcomings of these single mechanisms, using a combination of these mechanisms has been recommended for ensuring efficient and flexible cross-chain operations.

The hybrid mechanism can be superior to the single-mechanism scheme in terms of fault tolerance, wide application, and transaction time cost; however, its implementation process is relatively complex [23,24].

In recent years, cross-chain technology and its applications have been extensively studied. Wang et al. [25] introduced the concept of blockchain interoperability, designed a cross-chain operation model, and elucidated its application scenarios. Jing et al. [26] addressed regulatory challenges in blockchain and proposed a layered cross-chain regulatory architecture (CHA), along with a "regulatory chain-business chain" cross-chain collaboration model for regulatory governance. Hu & Xia [27] proposed a cross-chain transaction model that incorporates energy blockchain for carbon emission rights in microgrids to overcome the issue of excessive carbon emissions in traditional electricity trading. Ge et al. [28] addressed the problem of information silos across enterprises in the

existing seafood supply chain and proposed a water product trading model based on blockchain cross-chain technology to ensure secure transactions among enterprises. Overall, these studies have propelled the development of blockchain cross-chain technology and expanded its application domains.

From the aforementioned research, it is evident that blockchain technology holds significant value for applications in the field of copyright protection. Although blockchain cross-chain technology has gained increasing attention, there remains areas for improvement in this field currently, which can be summarized as follows.

- (1) While blockchain cross-chain technology exhibits excellent scalability, research on its integration with copyright protection is limited.
- (2) Digital resource copyright protection involves multiple stakeholders; however, research on how to enhance the efficiency of their participation is needed.

In light of the aforementioned issues, this study constructed a digital resource copyright protection solution based on blockchain cross-chain technology, striving to provide insightful considerations for the practical implementation of copyright protection for digital resources.

3. Theoretical basis and technical methods of research

Blockchain technology has evolved over time; however, initially, the blockchain was structured as a single chain. Given the inherent issues of isolated information and limited scalability associated with this model, a plethora of cross-chain techniques were explored, progressively shifting from single-chain enlargement to collaborative multi-chain development. The underlying principle of cross-chain operation is essentially the transfer of information or assets between chains supported by technological means, without inducing issues like value depreciation of blockchain or double-spending [29].

Due to the involvement of multiple stakeholders in these activities, the demand for cross-chain solutions in the sharing and copyright protection of digital resources is high. For instance, users not only aspire to share and obtain knowledge efficiently but also desire robust copyright protection for their intellectual contributions [30]. Earlier iterations of blockchain copyright safeguards were rooted in a single-chain format, which was constrained to a lone node on the chain, thereby processing only intra-chain tasks, which, in turn, impacted the facilitation of other operations. Conversely, cross-chain solutions in the blockchain realm offer pronounced benefits in terms of knowledge distribution and intellectual property defense, boasting superior business processing capacities per time unit, expedited processing velocities, and enhanced operational efficiency, among other advantages [31].

Given the prevailing cross-chain technological methodologies that include notary mechanisms, side chain/relay modalities, hash locking, distributed private key management, synthesis of notary mechanisms, and side chain techniques [32], the present study focused on a foundational technical methodology, that is, is the relay cross-chain model. It is a cross-chain technology that forwards and verifies transactions across different blockchains through relay nodes. It can achieve trustless cross-chain between different blockchains while ensuring the atomicity and reliability of cross-chain transactions. The basic architecture of a relay cross-chain model is illustrated in Fig. 1.

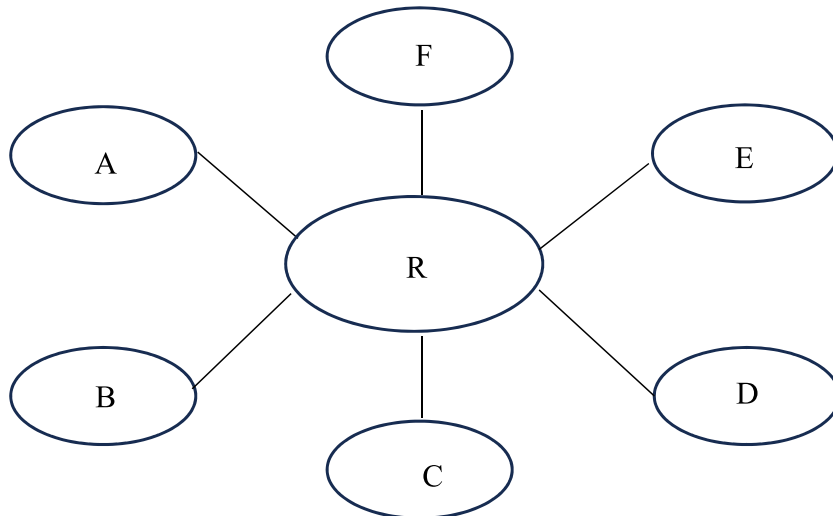


Fig. 1. Basic model of relay cross-chain.

4. Blockchain-based digital resource copyright protection scheme

4.1. Digital resource sharing activities and copyright protection scenarios

To protect the rights and interests of digital resources and copyright owners using blockchain technology, understanding the specific activities of displaying and sharing digital resources in the blockchain network is necessary.

Generally, the process of digital resource sharing involves the following steps: after identity authentication, the digital resource provider uploads the metadata of the created digital resource to the blockchain [33,34], while the digital resource itself is stored on other servers. The digital resource requester browses the metadata of the resource and requests access authorization from the resource provider when accessing the original resource. If approved, the desired digital resource can be obtained through the set extraction signature [35].

In the above process, scenarios related to digital resource copyright protection may include the following: publishing copyright-registered digital resources to the blockchain network, buying and selling digital resources with changed copyrights, and illegal use of creators' digital resources by others [36,37].

4.2. Specific scheme for the copyright protection of digital resources

Digital resource copyright management involves multiple entities, including copyright management departments, copyright arbitration departments, and users of digital resource sharing (providers and demanders of digital resources). To leverage the roles of various entities and improve the maintainability and scalability of the system, this study proposes a blockchain-based cross-chain digital resource copyright protection scheme, namely cross-chain copyright protection scheme (CCPS). CCPS considers the relay chain as the central chain, with resource sharing blockchain, copyright supervision blockchain, and copyright arbitration blockchain as parallel chains, and adopts a "relay chain parallel chain" approach to collaborate and perform the assigned task. The CCPS framework is displayed in Fig. 2.

In Fig. 2, R is the relay chain, representing the core of the entire cross-chain system, which aims to ensure system security and maintain normal system operation; A is the digital resource sharing blockchain, which involves users such as digital resource providers and demanders and forms node data for digital resource sharing and related operations under a certain consensus mechanism; B is the copyright supervision blockchain where the administrative management agency of copyright participates, and it is responsible for works such as copyright registration and management; and C is the copyright arbitration blockchain involving judicial institutions and is responsible for handling copyright protection work. The certification agency conducts identity authentication for each participating entity.

Adopting this cross-chain structure in "relay chain parallel chain" enables each parallel chain to handle closely related businesses. Each parallel chain transmits messages across the relay chain when information exchange is required [38].

CCPS has unique advantages and significance. First, considering that copyright protection in the display and sharing processes of digital resources is natural and crucial, using blockchain technology, CCPS can solve the problem of digital resource sharing and consider the copyright protection of resources. Second, multiple entities such as digital resource providers/users, copyright management departments, and arbitration institutions participate in copyright protection activities in the digital resource sharing process. Blockchain cross-chain technology is highly suitable for scenarios in which the participating entities are complex, and it can achieve digital asset transfer and information exchange across multiple blockchains [39,40].

Relevant business activities related to copyright protection in CCPS are displayed in Fig. 4.

As depicted in Fig. 3, the certification agency plays a role in verifying the identity of the copyright applicant, copyright

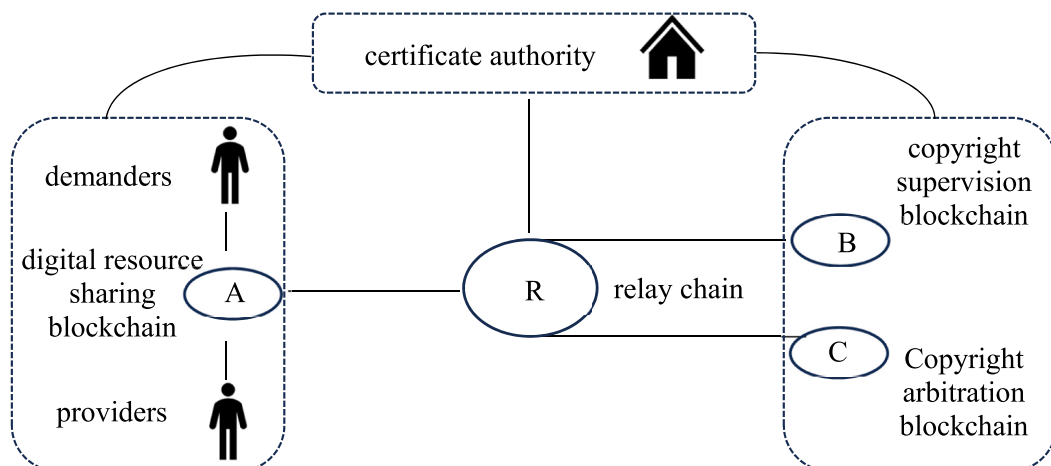


Fig. 2. Framework for cross-chain copyright protection of digital resources.

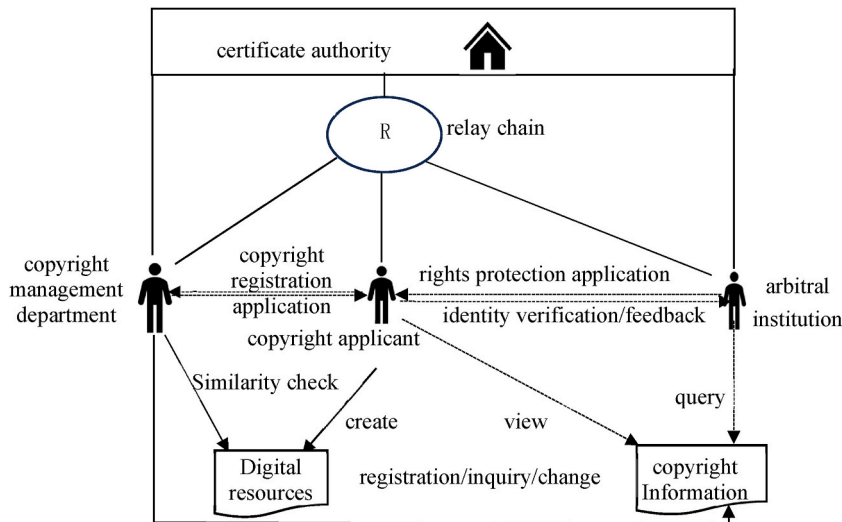


Fig. 3. Digital resource copyright protection-related services.

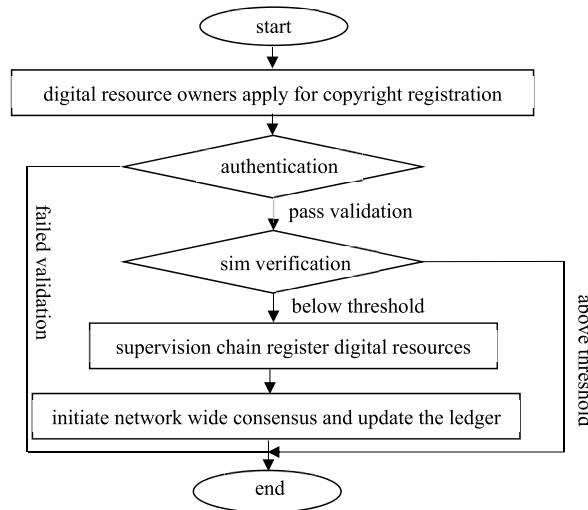


Fig. 4. Copyright registration process.

management agency, arbitration agency, and other entities. The copyright management department is responsible for accepting copyright registration applications from copyright applicants, checking the similarity of digital resources during copyright registration and querying and changing copyrights. Among them, the similarity check adopts the modified SimHash algorithm [41]. The arbitration institutions are responsible for handling copyright owners' rights protection applications, querying copyright information of digital resources, and providing feedback on rights protection applications.

In Fig. 4, the dashed line between the two entities indicates related business transactions between the two entities; these entities do not directly contact each other, and information exchange between them occurs through a relay chain.

4.3. Digital resource copyright management business process

4.3.1. Copyright registration

Copyright supervision blockchain can achieve copyright registration of digital resources, and its specific process is as follows.

- (1) A threshold for determining originality is set, which indicates the highest degree of similarity between the digital resource to be registered and existing resources.
- (2) The digital resource provider applies for copyright registration to the copyright supervision blockchain node through the relay chain. After the consent is obtained, a smart contract is initiated for the identity verification and digital resource similarity

verification of the applicant. When conducting similarity checks, the blockchain calls the off-chain storage resource database to compare the similarity with the current resources applying for copyright registration and obtain the similarity value.

- (3) The obtained similarity value is compared with the pre-set originality threshold. The similarity value higher than the originality threshold indicates that the digital resource lacks originality, and the copyright registration application is rejected. If the copyright registration application is successful, the hash value of the digital resource and the applicant's identity information are stored in the copyright supervision blockchain, as displayed in Fig. 4.

4.3.2. Resource trading and copyright changes

Transaction and transfer of intellectual property are realized as follows. When users wish to purchase digital resources whose owner is unknown, they can search through the copyright supervision blockchain. Resource trading triggers a copyright trading smart contract to transfer transaction amounts and change copyright registration information (Fig. 5).

4.3.3. Copyright protection

The copyright arbitration blockchain handles infringement behaviors that occur during the sharing, trading, or transfer of digital resources. Upon discovering infringement, the digital resource copyright owner submits the information of the infringer and the infringing resource to the copyright arbitration blockchain through the relay chain. After verifying the identity of the applicant, the arbitration department verifies whether infringement has occurred through the relay chain to the digital resource sharing blockchain and copyright supervision blockchain. Once verified, the infringement processing smart contract is triggered, which determines the degree of infringement, settles the compensation amount, initiates a nationwide consensus, and updates the copyright registration ledger. The specific process is displayed in Fig. 6.

4.4. Analysis of digital resource copyright protection schemes

CCPS exhibits excellent operability, scientific rationality, and high computational performance. Its advantages can be summarized as follows:

First, CCPS can ensure that the digital resources on the chain exhibit excellent originality. Comparing resources with various offline platform resources or databases is critical before uploading digital resources to the blockchain. If the comparison result shows that the resources to be uploaded exhibit remarkable similarity to the existing resources, then the resources cannot be uploaded.

Second, when digital resources exhibit high originality and can be linked, only the metadata or index information of the resources should be uploaded to the chain, and the complete digital resources should be uploaded to a trusted cloud server [42]. This process saves storage space on the blockchain and improves the availability of copyright protection schemes.

Third, CCPS is based on relay and parallel chains such as copyright supervision and arbitration. In case the integration of more businesses is needed in the future, only corresponding parallel chains should be added, ensuring that the cross-chain solution exhibits excellent scalability and improving its availability.

Finally, from the transaction throughput perspective, CCPS involves multiple blockchains that are interconnected but independent. The number of transactions per second generated by these blockchains is considerably higher than that generated by a single blockchain, which effectively solves the problem of low throughput for copyright protection transactions.

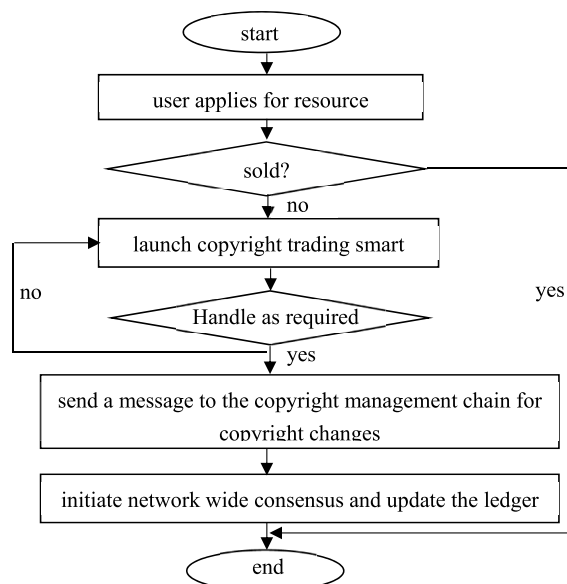


Fig. 5. Copyright trading process.

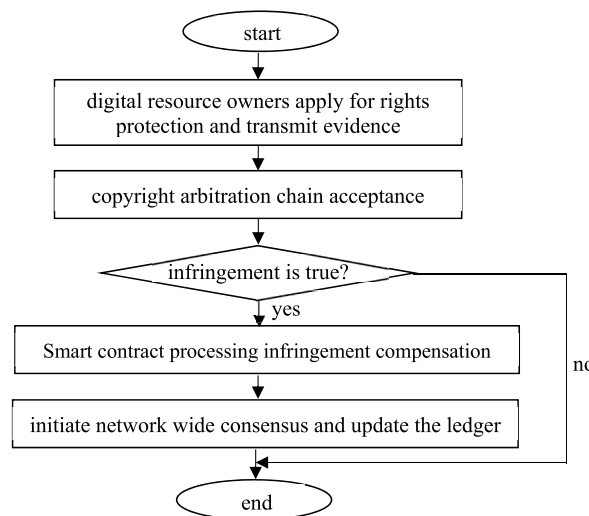


Fig. 6. Copyright protection process.

5. Conclusion and insights

5.1. Conclusion

This article considers digital resources as a shareable resource and addresses problems related to copyright registration, copyright changes, and copyright protection during the sharing process. Blockchain technology exhibits distribution, tamper resistance, traceability, and multi-party maintenance, which can be applied for the copyright protection of digital resources. However, the blockchain single-chain structure requires each entity node on the chain to back up all business data of the industry on the chain (regardless of whether it is related to itself or not), rendering it difficult for each entity to perform its respective duties effectively. In this study, a digital resource copyright protection scheme is proposed based on blockchain cross-chain, which is suitable for scenarios involving the participation of multiple entities in resource sharing and copyright protection. The proposed model ensures the blockchain system is maintainable and scalable.

This study provides theoretical support for the interoperability between blockchains representing various operations in the process of digital resource sharing and copyright protection. Additionally, the study introduces new ideas for the mechanism of digital resource copyright protection. Furthermore, this solution extends the application scope of blockchain technology and offers practical guidance for the implementation of digital resource copyright protection.

5.2. Research insights

5.2.1. Emphasize copyright protection for digital resources

In the digital age, copyright protection for digital resources is of paramount importance. With the rapid development of the Internet and technology, the dissemination and sharing of digital resources have become integral in daily life. However, this has also led to issues of piracy, infringement, and unauthorized use. To safeguard intellectual property and encourage innovation, it is crucial to strengthen copyright protection for digital resources. First, copyright protection for digital resources helps safeguard intellectual property rights. Authors, artists, creators, and inventors invest significant time and effort in creating digital resources, and they should have legitimate rights to control and benefit from their works. By ensuring the legality of copyrights, we can encourage more individuals to engage in the creation and sharing of digital resources, thereby promoting innovation and cultural diversity. Second, copyright protection contributes to maintaining the quality and sustainability of digital resources. The protection of intellectual property incentivizes creators to invest more resources and time in improving their works and enhancing their quality. Simultaneously, it provides economic incentives for the continuous creation and updating of digital resources, which helps sustain health of the digital ecosystem. Furthermore, copyright protection for digital resources helps protect the rights of users. Legally acquired digital resources are typically protected by copyright laws, meaning that users can enjoy high-quality content without worrying about infringement or fraudulent activities. Additionally, copyright protection aids in combating piracy and infringement, thereby maintaining the security and integrity of the digital environment.

Prioritizing copyright protection for digital resources is a crucial step in safeguarding intellectual property, promoting innovation, protecting user rights, and driving the growth of digital economy. It is crucial to ensure that copyright for digital resources is fully respected and protected through strengthened legal frameworks, technological means, and educational awareness, making the digital age more prosperous and sustainable.

5.2.2. Vigorously develop blockchain cross-chain technology

Blockchain cross-chain technology holds immense potential and offers multiple advantages that can accelerate the development of digital economy. Cross-chain technology aims to address the isolation and interoperability issues between different blockchain networks, enabling them to collaborate and communicate more closely. The cross-chain technology can enhance the interoperability of blockchain networks. Currently, different blockchain networks often operate in isolation, making it challenging for them to interact with each other. This leads to difficulties in the transfer of digital assets, limitations of smart contracts, and data isolation. Through cross-chain technology, different blockchains can exchange data and value more freely, promoting the globalization of digital economy. Furthermore, cross-chain technology can improve the liquidity of digital assets. Digital assets, such as cryptocurrencies and tokens, are often stored on different blockchains. Cross-chain technology allows users to easily transfer and exchange digital assets between different chains, increasing the liquidity and flexibility of assets. This provides investors and users with more choices and reduces the complexity of digital asset management. Furthermore, cross-chain technology contributes to digital identity management. It can offer more secure and decentralized digital identity solutions, enhancing user authentication and privacy protection. This is crucial for strengthening online security and safeguarding user data.

To summarize, vigorous development of the blockchain cross-chain technology can significantly benefit the digital economy by enhancing interoperability, improving asset liquidity, and enhancing digital identity management. It is essential to invest in and advance this technology to unlock its full potential for the growth and development of digital economy.

5.2.3. Actively apply blockchain cross-chain technology for copyright protection

Blockchain cross-chain technology can be introduced to overcome challenges associated with single-chain or consortium-chain approaches for copyright protection such as low system processing efficiency, poor scalability, and misalignment with real-world transactions. A copyright protection solution based on blockchain cross-chain technology offers several advantages. First, it enables various stakeholders in copyright protection to better perform their respective roles, thus improving work efficiency and system scalability. Second, it can establish a more efficient, transparent, decentralized, and global copyright ecosystem. Participants can better fulfill their roles, achieving the goals of copyright protection. Additionally, it provides more opportunities and safeguards for digital resource creators, copyright management organizations, distribution platforms, and end-users.

Given these advantages, stakeholders in blockchain technology should strive to promote the development of blockchain technology, especially cross-chain technology, to make blockchain technology more mature and stable. They should also actively expand the application areas of blockchain technology, harnessing the advantages of blockchain and its cross-chain technology to provide efficient, secure, and global solutions for copyright protection and various other domains.

Certain limitations exist in this study. For example, although the study proposes a copyright protection scheme for digital resources, its implementation remains to be validated. Additionally, the study analyzed the proposed scheme only qualitatively, lacking quantitative analysis based on experimental data. Future research can focus on the following aspects: First, implementing a cross-chain-based copyright protection solution in code and analyzing the advantages of cross-chain systems through experimental data. Second, to achieve interoperability across different blockchains, there is a need to establish common cross-chain standards for ensuring seamless collaboration among various blockchain platforms. Furthermore, privacy and security issues remain critical challenges in cross-chain interactions. To ensure the security of cross-chain transactions and user privacy, future research should concentrate on developing robust privacy protection technologies and security mechanisms.

Data availability statement

No data were used for the research described in the article.

CRedit authorship contribution statement

Renqiang Xie: Writing – original draft, Resources, Methodology, Conceptualization. **Min Tang:** Writing – review & editing, Investigation.

Declaration of competing interest

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